Netflix Movies Analysis using Python (Pandas)

Project Overview

This project analyzes the Netflix Movies and TV Shows dataset using Python and Pandas to uncover trends in Netflix's content library. The project follows the complete data analysis lifecycle, including data cleaning, exploratory data analysis (EDA), visualization, and business insights.

Objectives

The primary objectives of this project are:

- Clean and preprocess the Netflix dataset.
- Explore the structure and quality of the data.
- Analyze trends in Netflix Movies and TV Shows.
- Identify the most common genres, ratings, countries, and directors.
- Generate business insights through data visualization.

Dataset

- **Dataset Name:** Netflix Movies and TV Shows
- **Source:** Kaggle
- **Format:** CSV
- **Records:** 8,807
- **Columns:** 12

Features

- Show ID
- Type
- Title
- Director
- Cast
- Country
- Date Added

- Release Year
- Rating
- Duration
- Listed In (Genre)
- Description

🛠️ Tools & Technologies

- Python
- Pandas
- Matplotlib
- Visual Studio Code
- Git
- GitHub

📁 Project Structure

```
Netflix-Movies-Analysis/
│
├── dataset/
│   ├── netflix_titles.csv
│   └── netflix_titles_cleaned.csv
│
├── notebooks/
│   └── netflix_analysis.ipynb
│
├── scripts/
│   ├── cleaning.py
│   └── eda.py
│
├── images/
│   ├── movies_vs_tvshows.png
│   ├── top_genres.png
│   ├── top_countries.png
│   ├── release_year.png
│   ├── ratings.png
│   └── directors.png
│
├── report/
│   └── Netflix_Movies_Analysis_Report.pdf
│
├── README.md
├── requirements.txt
└── LICENSE
```

🧹 Data Cleaning

The following preprocessing steps were performed:

- Removed duplicate records.
- Filled missing values in:
  - Director
  - Cast
  - Country
  - Rating
  - Duration
  - Date Added
- Converted **Date Added** to datetime format.
- Removed leading and trailing spaces.
- Saved the cleaned dataset.

📊 Exploratory Data Analysis (EDA)

The following analyses were conducted:

- Movies vs TV Shows
- Top 10 Genres
- Top 10 Countries
- Top 10 Directors
- Content Rating Distribution
- Content Released by Year

📈 Key Findings

1. Movies dominate Netflix.

Netflix contains significantly more Movies than TV Shows.

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2. United States has the highest number of titles.

The United States contributes the largest amount of content, followed by India and the United Kingdom.

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3. Netflix expanded rapidly after 2015.

The number of releases increased substantially after 2015, reflecting Netflix's growth in original and licensed content.

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4. Drama is the most common genre.

Drama, International Movies, and Comedy are among the most frequent genres.

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5. TV-MA is the most common rating.

Most Netflix titles target mature audiences.

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6. Some directors have multiple titles.

A small number of directors appear repeatedly, while many contribute only one title.

💼 Business Recommendations

Based on the analysis, Netflix could:

- Invest more in popular genres such as Drama and International Movies.
- Expand production in high-performing countries.
- Increase family-friendly content to broaden its audience.
- Improve recommendation systems using genre, rating, and release year.
- Use historical viewing trends to guide future content investments.

📷 Visualizations

Include screenshots of your generated charts in the `images/` folder, such as:

- Movies vs TV Shows
- Top Genres
- Top Countries
- Rating Distribution
- Release Year Trend
- Top Directors

These visuals make your GitHub repository more engaging.

📌 Conclusion

This project demonstrates a complete data analysis workflow using Python. Through data cleaning, exploration, visualization, and interpretation, meaningful insights about Netflix's content library were identified. The project highlights the importance of preparing data before analysis and using visualizations to support data-driven decision-making.

🚀 Future Improvements

- Build an interactive dashboard using Power BI.
- Integrate IMDb ratings for richer analysis.
- Perform sentiment analysis on movie descriptions.
- Develop a machine learning model to predict content popularity.
- Deploy an interactive dashboard using Streamlit.